

The Black Brant V rocket motor produces an average thrust level of 15,596 pounds and an action time of 32.42 seconds. The primary diameter of the Black Brant V is 17.26 inches and it is 210 inches long. Loaded weight of the motor including hardware is 2,803 pounds which includes 2243 pounds of propellant.

https://sites.wff.nasa.gov/code810/vehicles/Black_Brant_V.pdf

Assuming a payload of 1000 pounds, a drag coefficient of 0.75, and a launch angle with respect to the horizontal of 45 degrees, how far away will the rocket be when it hits the ground?

You can neglect changes in thrust with changing atmospheric pressure and the curvature of the earth, but need to account for changes in atmospheric density with elevation. You may need to account for changes in the value of g if the rocket goes high enough.

Write the MATLAB code to solve this problem. Define all the necessary initial condition values to solve this problem.

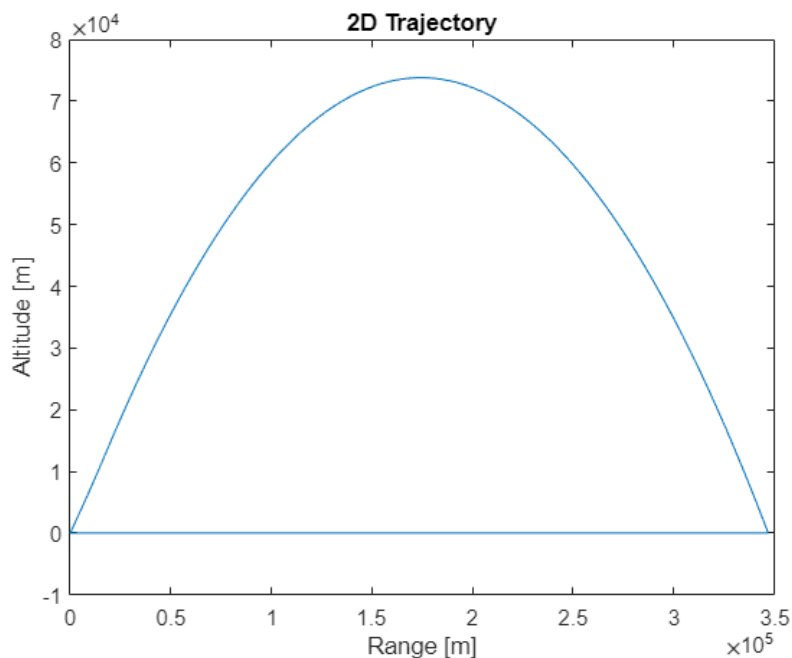
To complete this objective a valid solution technique must be used and final answer must be within $\pm 1\%$ of the correct answer.

You may discuss this problem with your classmates, but each person must write and submit their own MATLAB code. Codes will be compared using the Turnitin software.

Answer:

1. Range = 346.942 km

include a plot of your 2D trajectory:



```
clc
clf
close all
clear
```

%aeroDataPackage is the atmospheric and gravitational data toolbox I used

```
F_thrust_mag = convforce(15596, 'lbf', 'N'); % N
ActionTime = 32.42; %seconds
D_vehicle = convlength(17.260, 'in', 'm'); %m
L_vehicle = convlength(210, 'in', 'm'); %m
M_payload = 453.5924; %kg
M_hardware = 254.0117; %kg
M_fuel = 1017.408; %kg
M_total_val = M_payload + M_hardware + M_fuel; %kg
```

```
A_vehicle = (pi/4)*D_vehicle^2; %m^2
```

```
C_d = 0.75;
```

```
theta_launch = deg2rad(45); %rads
```

```
m_dot = M_fuel/ActionTime; %kg/sec
```

```
delta_t = 0.001; %seconds IMPORTANT
```

```
size = 1000000;
```

```
gz = zeros(1,size);
x = zeros(1,size);
y = zeros(1,size);
V_x = zeros(1,size);
V_y = zeros(1,size);
Theta = zeros(1,size);
Rho = zeros(1,size);
F_thrust = zeros(1,size);
F_thrust(1) = F_thrust_mag;
F_drag = zeros(1,size);
speed = zeros(1,size);
F_x = zeros(1,size);
F_y = zeros(1,size);
accel_x = zeros(1,size);
accel_y = zeros(1,size);
M_total = zeros(1,size);
```

```

M_total(1) = M_total_val;

Theta(1) = theta_launch;

x(1) = 0; %m
y(1) = 0; %m
V_x(1) = 0; %m/s
V_y(1) = 0; %m/s

g = 9.81; %m/s

F_thrust_x = cos(theta_launch)*F_thrust(1);
F_thrust_y = sin(theta_launch)*F_thrust(1);

accel_x(1) = (F_thrust_x-F_drag(1))/M_total(1);
accel_y(1) = ((F_thrust_y-F_drag(1))/M_total(1))-g;

ii = 1;
% Calculations
while y(ii) >= 0

    % Atmospheric Properties
    [T, a, P, rho] = atmosisa(y(ii));
    Rho(ii) = rho; % kg/m^3

    gz(ii) = gravityzonal([y(ii)+6378.1363e3 0 0]); %Aerospace toolbox gravity tool, gravity
    does change more than 1%

    ii = ii+1;

    if (ii*delta_t) <= ActionTime

        F_thrust(ii) = F_thrust_mag;
        M_total(ii) = M_total_val - m_dot * (ii-1) * delta_t;
        Theta(ii) = theta_launch;

    else

        M_total(ii) = M_hardware + M_payload;
        F_thrust(ii) = 0;

    end

```

```

% Drag force calculation
F_drag(ii)= 0.5*C_d*Rho(ii)*A_vehicle*(V_x(ii-1)^2+V_y(ii-1)^2);

% Sum of forces calculations
F_x(ii)= F_thrust(ii)*cos(Theta(ii-1))-F_drag(ii)*cos(Theta(ii-1));

F_y(ii)= F_thrust(ii)*sin(Theta(ii-1))-F_drag(ii)*sin(Theta(ii-1));

% Accelerations
accel_x(ii)= F_x(ii)/M_total(ii);
accel_y(ii)= (F_y(ii)/M_total(ii))+gz(ii-1);

% Velocity
V_x(ii)= V_x(ii-1)+accel_x(ii)*delta_t;
V_y(ii)= V_y(ii-1)+accel_y(ii)*delta_t;

speed(ii) = (V_x(ii)^2 + V_y(ii)^2);

% Position
x(ii)= x(ii-1) + V_x(ii) * delta_t;
y(ii)= y(ii-1) + V_y(ii) * delta_t;

% angle
if V_x(ii) ~= 0 && (ii*delta_t) > ActionTime
%I think this section is wrong but I dont know how to fix this assumption
    Theta(ii)= atan2(V_y(ii), V_x(ii));
elseif (ii*delta_t) > ActionTime
    Theta(ii) = pi/2;
end

end

max_y = max(y)
max_y = 7.3820e+04
fprintf("Final Altitude: %.6f m", y(ii))
Final Altitude: -0.758602 m
fprintf("Range: %.6f m", x(ii))
Range: 346941.676665 m
plot(x,y)
title("2D Trajectory")
xlabel('Range [m]')
ylabel('Altitude [m]')

```